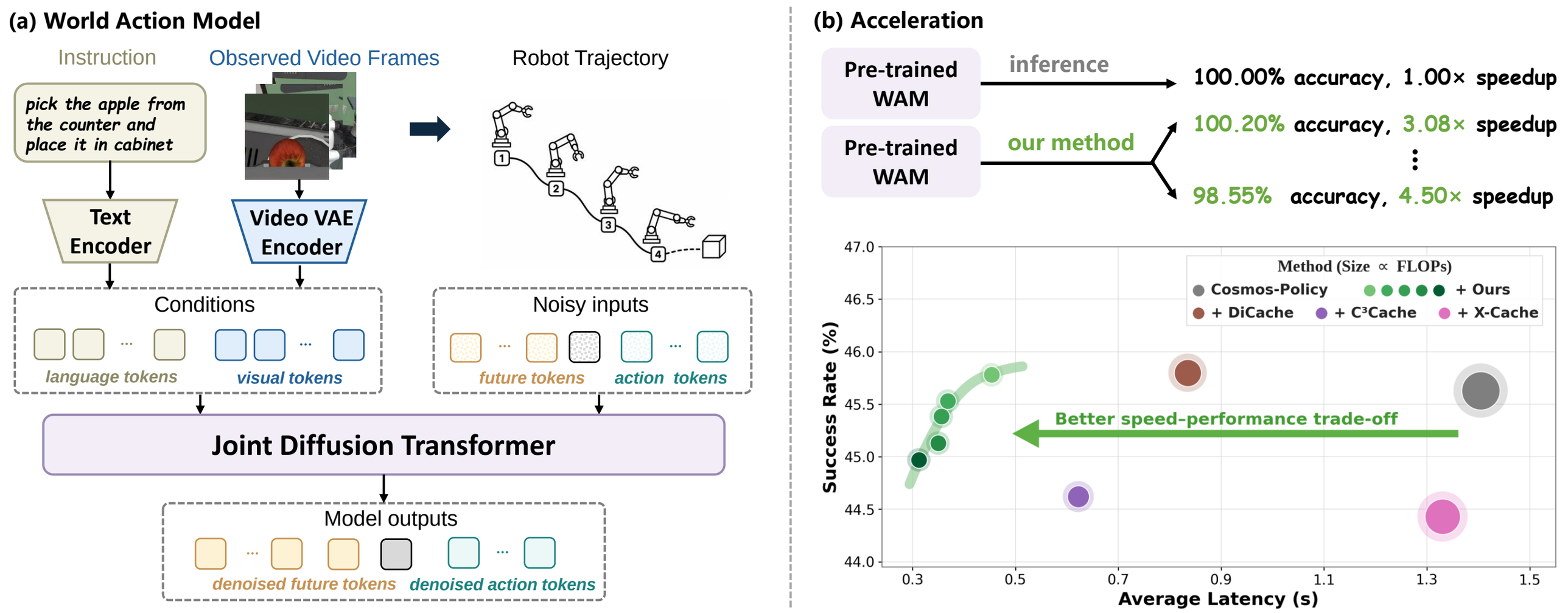

(a) World Action Model
Instruction
Observed Video Frames
Robot Trajectory
pick the apple from the counter and place it in cabinet
Text Encoder
Video VAE Encoder
Conditions
language tokens
visual tokens
Noisy inputs
future tokens
action tokens
Joint Diffusion Transformer
Model outputs
denoised future tokens
denoised action tokens
(b) Acceleration
Pre-trained WAM
inference
100.00% accuracy, 1.00× speedup
Pre-trained WAM
our method
100.20% accuracy, 3.08× speedup
98.55% accuracy, 4.50× speedup
Method (Size ∝ FLOPs)
Cosmos-Policy
+ Ours
+ DiCache
+ C³Cache
+ X-Cache
Better speed-performance trade-off
Success Rate (%)
Average Latency (s)
47.0
46.5
46.0
45.5
45.0
44.5
44.0
0.3
0.5
0.7
0.9
1.1
1.3
1.5